

NETFLIX CONTENT TYPE PREDICTION USING MACHINE LEARNING

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PROJECT ABSTRACT

The rapid growth of digital streaming platforms has increased the need for intelligent content classification systems. This project aims to develop a machine learning based application capable of predicting whether a Netflix title belongs to the category of Movie or TV Show using content-related attributes. The Netflix Titles Dataset was used for training and testing the model. The dataset contains information such as director, cast, country, release year, rating, listed category, month added, year added, duration in minutes, and number of seasons. Data preprocessing techniques including handling missing values and label encoding were applied to prepare the dataset for model training.

A Random Forest Classifier was trained to identify patterns within the dataset and accurately classify content types. The trained model was integrated into a Flask web application that allows users to enter content details and obtain instant predictions. The application accepts ten input features and processes them through the trained model to determine whether the content is a Movie or a TV Show. This project demonstrates the practical implementation of machine learning, data preprocessing, model deployment, and web application development, providing an efficient solution for Netflix content categorization.